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Tagged

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Strengths of this report include some quantitative analysis (see Appendix B for some of the data) and good justifications in Section 3.3.

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1 High-Level Description of Project

1.1 Motivation

Laser tag was released to the general populace 35 years ago, and has had minimal changes with respect to portability of the game, bulk of the equipment and availability for personal use [1]. Laser tag is a recreational activity that relies on a handheld device, often in the form of a mock firearm that emits infrared beams at targets. These targets are typically worn by players or a part of the play area. Upon tagging these targets, points are scored, tallied and used to determine a winner at the end of a session.

Initially developed for combat training, laser tag has long been limited to the indoor arena experience, involving heavy equipment and incredibly specific targets. The indoor laser tag games have 2 core components - the gun and the vest - with the weight generally ranging from 5.5 to 10 pounds [2], and only 4 main areas for embedded IR sensors on the vest, which cover little of the overall area. More recently, it has become common for laser tag companies to provide outdoor laser tag games.

With the outdoor experience, however, comes further target limitations. In most outdoor laser tag equipment, there are only one to two sensors, with one being located on the gun, as seen in **Figure 1** below, and the other commonly mounted on a headband worn by the player. This detracts from the initial purpose of laser tag - to simulate tag - as you can only tag another player in incredibly specific locations that are either on a small area of the body, or not a part of the body at all. Both the indoor and outdoor laser tag experiences are also generally not for personal use, and require a third party to complete setup, and provide equipment.



Figure 1: Outdoor Laser Tag Gun Schematic [3]

This project, Tagged, aims to redesign and bring the key elements of laser tag to players everywhere. Tagged is to laser tag as the video game console is to the gaming arcade; offering portability and convenience for users to start a game anywhere, anytime.

1.2 Project Objective

The aim of Tagged is to design a portable gameplay system that brings the full laser tag experience to the user, is easy to setup and allows users to play the game in any arena of their choosing.

1.3 Block Diagram

The design project consists of three major components that the group will be designing: the laser-firing glove, a vest and a central base station. These components have been laid out in **Figure 2** to provide a pictorial view of the inner workings of the system.

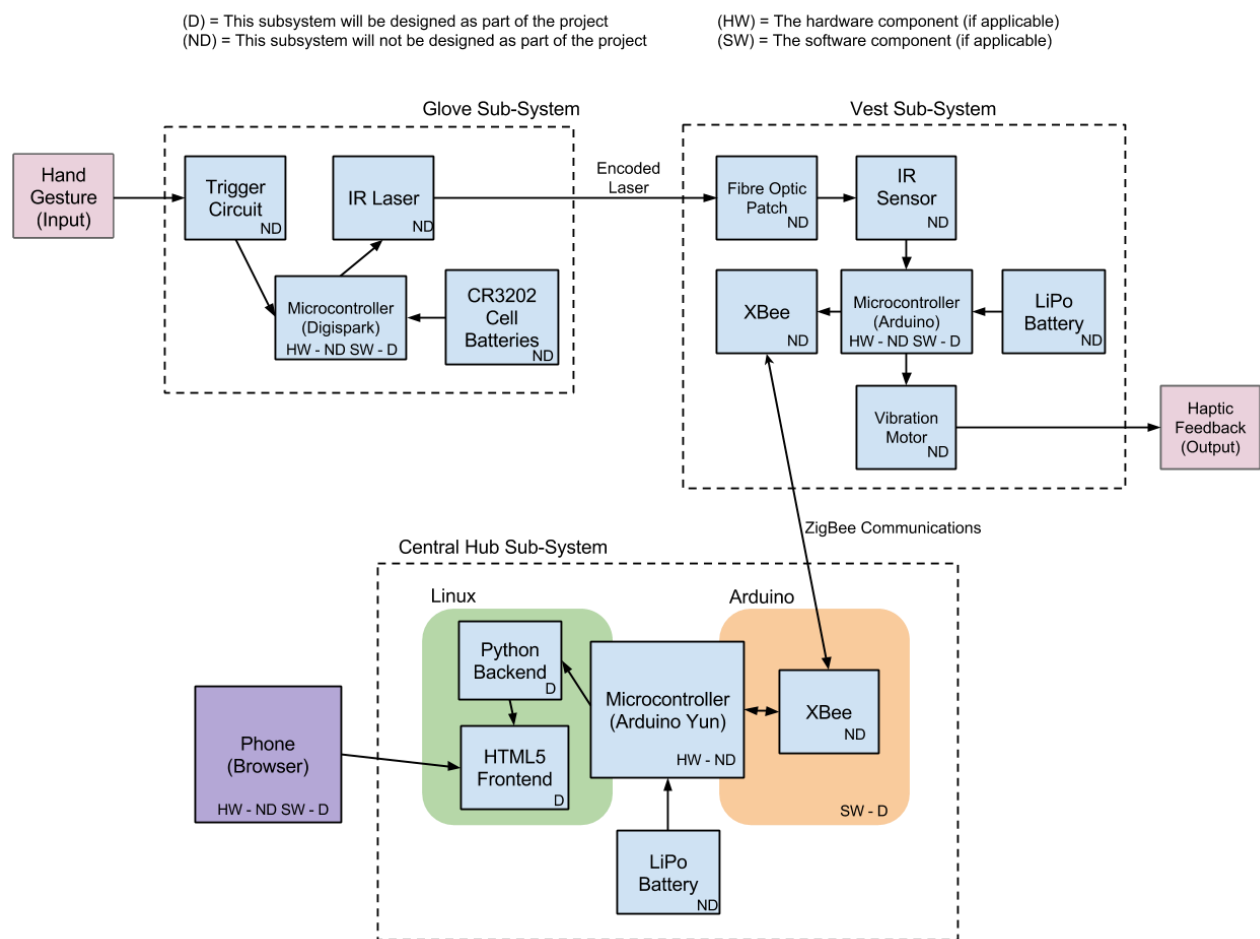


Figure 2: Block Diagram of Tagged System

The project will replace the typical firearm form-factor with a gesture-sensing glove to shoot pulses of IR beams. A slimmed down vest does away with the bulk of the classic indoor laser tag sensor rig, while a

base station will compute real-time scoring, and provide an arena in the form of a network which the vests will use for communication.

As the block diagram in **Figure 2** illustrates, the majority of the electrical components that make up the glove will not be designed by the group, but on the software end, design will be required to filter signals from the sensors. The software will be hosted on a Digispark, a quarter-sized Arduino board, which is ideal as it reduces component real estate on the glove. The circuitry involved in triggering the laser to shoot will be either connected or disconnected depending on the contact between patches of conductive fabric. If the circuit is connected, then a shot from the laser will be fired - otherwise, the laser diode will remain off. It should be noted that the glove is a self contained device that will not require communication with the vest or base station to function.

The vest mimics the same form as those typically found at indoor laser tag establishments, but will lack the heavy sensor pods. Instead, sensors will be placed to cover areas of the stomach, upper back, and upper torso, providing users with increased target areas. It is expected that the vest will require extensive design work to create a low bulk IR sensor. The proposed approach is to fashion fibre optic fabric into a patch where all the fibre optic ends feed into an IR sensor. The patches will then be sewn into the vest to act as target areas. The signals from the IR sensors will be sent to the microcontroller, which, like the glove subsystem, will perform signal processing on the software side. If a valid hit is detected, a signal is sent through the onboard Xbee module to the base station via ZigBee communications. There will be some additional design work that goes into determining component placement and attachment to achieve proper enclosure of electrical components.

The base station will act as a central hub for all vests to communicate with, by establishing a network through the Xbee module. The bulk of design for this sub-system will be in the software. In particular, the software will handle logging scores and enforcing gameplay configurations. If time permits, the project would like to extend the base station feature to providing a web interface for users to modify gameplay settings. The base station uses an Arduino Yún which provides a Linux platform as well as the original Arduino environment. The dual platform serves all the project's functional requirements well; the Arduino environment can handle the hardware interfacing while the Linux half allows for web development to be hosted on the microcontroller. An onboard WiFi antenna mitigates the need to find additional hardware integration for users to access the base station's hosted web pages.

2 Project Specifications

2.1 Functional Specifications

To help roadmap the project, the three main components and their essential functions break down are displayed in Table 1.

Table 1: Essential Function Specifications of the Tagged System

Component	Description
Glove	
Gesture Recognition	The sub-system shall detect when a specific trigger action has been performed with an accuracy of at least 90%, and then subsequently fire off an IR laser pulse.
Vest	
Infrared Detection	The vest should be able to detect an IR pulse fired from the glove's laser at least 70% of the times it has been hit.
Haptic Feedback	Upon being hit, the vest shall provide haptic feedback to signal to the wearer that they have been tagged.
Communication	The vest should also be able to communicate with the base station to update the score upon being hit.
Base Station	
Communication	The base station should be able to create a network over which communications from the vest shall be received.
Range	The range of the base station should be at least 85 feet in a direct line-of-sight scenario.
Game Enforcement	The base station should be capable of timing the game, and communicating to the vest to induce vibrations signaling to the players that the game is either starting or stopping.
Score Keeping	The base station should be able to keep track of each player's score for a game of minimum 2 players.

As well, nice-to-have features that could be implemented are listed in Table 2 which would take the Tagged project beyond a satisfactory state of completion.

Table 2: Non-Essential Functional Specifications for the Tagged System

Component	Description
Phone App	The user should be able to access the in-game stats and manage game settings through a web interface to the base station.
Gameplay Management	Users should be able to manipulate game settings from the base station, prior to the start of each game. The controllable setting will include setting game duration, and whether scoring will be calculated for a team setting or as individual.
Extended Gameplay Options	At least one extended gameplay mode should be coded for, with the corresponding algorithm in place for determining the score. This will be an option to be selected from the base station.

2.2 Non-Functional Specifications

In addition to outlining the functional specifications of Tagged, the group will be designing the system with the following characteristics in mind, which have been listed as non-functional specifications in Tables 3 and 4 as follows.

Table 3: Essential Non-Functional Specifications for the Tagged System

Component	Description
Safety	All electronic components must be properly enclosed - as per ESA standards. Safety instructions shall also be provided regarding laser use. No sharp or jagged edges will protrude from the vest or glove.
Battery Life	All components of the system must be able to sustain themselves for a minimum of 1.5 hours before needing a recharge.
Portable	All system components should be able to comfortably fit within a compact sized suitcase [4], with at most a height of 63 centimetres, width of 36 centimetres, and depth of 21 centimetres.
Comfort	Game equipment will be primarily composed of soft, flexible material, and will accommodate a range of US standard small, medium and large sizes. See appendix B for details.

A list of non-essential non-functional specifications, if met, which would further strengthen the niche Tagged is trying to fill are listed in Table 4.

Table 4: Non-Essential Non-Functional Specifications for the Tagged System

Component	Description
Lightweight	The total weight of the glove and vest combined must be under 7 pounds.
Durable/Rugged	The vest should be capable of withstanding being moved 1 meter across a brick surface with a 10 pound weight placed on top. No abrasions should form.
Target Coverage	The fibre optic patches should cover at least 30% of the total area of the vest.

3 Detailed Design

3.1 Glove Subsystem Design

Given the project's substitution of the traditional laser gun for a smart glove, the typical trigger mechanism used is not used. Instead, the glove should respond to the user's hand position in order to fire off a laser pulse in the direction pointed to by the index finger. The primary focus for the glove's design can be summarized into the following major points:

- The glove comfortably fits the user's hand
- The glove is primarily made of soft material
- The glove can record and interpret the user's hand positions
- The glove is able to shoot a laser pulse, if it detects the appropriate hand position
- The glove will be powered by a coin cell battery to keep it slim

3.1.1 First Iteration

The initial design leverages the use of flex sensors mounted along the thumb, the index and the ring finger as the trigger mechanism. These sensors are crucial in detecting when certain fingers are curled or straightened. The two most common "shooting" hand gestures, as shown in **Figure 3**, would have the ring finger curled and the index straightened. A flex sensor that monitors the thumb determines when the player has "triggered" a shot by bending the thumb.

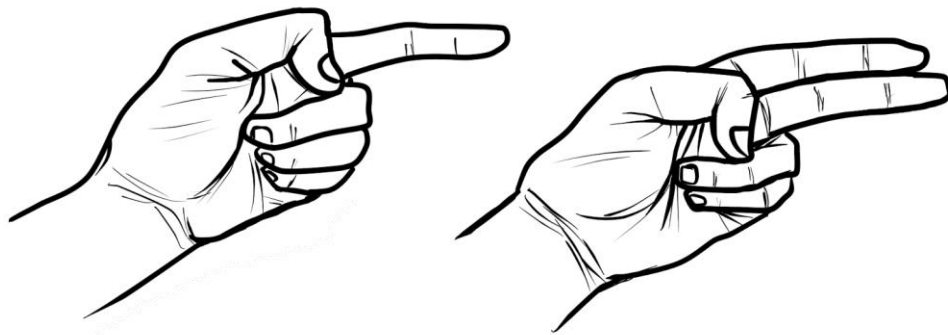


Figure 3: Two Common "Shooting" Hand Gestures

The flex sensors are encased by a layer of plastic as a measure of protection, before fixing it to the glove base. The glove base is a form-fitting, breathable nylon glove commonly used for sporting and recreational purposes. A Digispark microcontroller mounted on the glove performs signal filtering on the input from the flex sensors. Together, the Digispark and the flex sensors form the gesture-sensing component of the glove for the Tagged laser tag system.

Due to the unique nuances of an individual's hand, the glove system needs to be calibrated so that the Digispark microcontroller has a baseline measurement to filter input signals against. Once calibrated, the system will be able to more accurately interpret the gestures being made by the wearer. In particular, calibration requires a calibration sequence of the wearer's hand relaxed, straightened and curled into a fist to better adapt to the wearer's naturally inclined movements.

Upon some early testing with the flex sensors, the flex sensors yielded consistent results but had an observably slow response time. Additionally, mounting them to the user's fingers detracted from their ability to make smooth natural movements. As such, the use of flex sensors was deemed unusable for the project use.

3.1.3 Second Iteration

The glove's laser pulse triggering system requires an alternative to flex sensors, while maintaining the point-and-shoot concept of the Tagged system. After further brainstorming, the group had to rethink what a player expects out of Tagged and how to enhance the user's experience of laser tag. The result was to try capacitive or touch-related sensors.



Figure 4: Gesture that Triggers Laser Pulse

Another common gesture that emulates shooting would be to steady the aiming hand with the other as shown in **Figure 4**. Using this gesture as a starting point, the new design aims to only trigger a laser pulse through very deliberate actions. Instead of relying on capturing the resistance values of three flex sensors, processing the resultant data, and then firing off a laser beam, the new design works off the notion of 'completing a circuit'.

In particular, there are specific areas on the gloves made from the conductive fabric which will need to come into contact with each other in order for the laser to fire. By simply 'completing the circuit' - that is, pressing the appropriate parts of the user's gloves together - the laser will fire. This

approach minimizes the amount of latency needed to shoot the laser on the glove, as compared to the previous design iteration. It requires no analysis of data, and merely identifies that all the requirements to fire have been met, in which case the laser will fire. Additionally, it gets rid of the need to calibrate, as there are no flex sensors with resistance thresholds that need to be established.

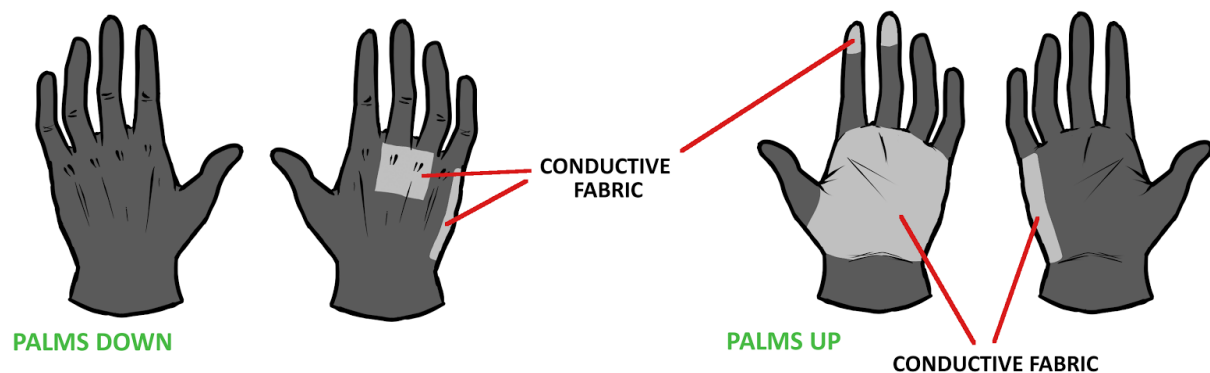


Figure 5: Both gloves with their respective conductive zones

As illustrated in **Figure 5**, above, the gloves will not be symmetrical with respect to the conductive spots on them. This implies that the prototype will be designed for right hand users. In order to fire a shot, the right hand of the user must be cupped by their left hand, such that the side of the user's right palm furthest away from their thumb makes contact with the palm of their left hand. This is not the only mechanism the system will rely on to detect a shot, and is rather the *shooting stance position* (see **Figure 4**). In order to actually trigger the laser and fire a shot, once the user's right hand is cupped by their left, they can simply use the fingers on their left hand with conductive fabric tips to tap the one-by-one inch patch of conductive fabric situated by the middle finger knuckle on the back of the user's right hand (see **Figure 5**).

3.1.4 Communication and Data Encoding

To keep track of scoring, the laser pulse sent by each glove is an encoded pulse modulated beam that contains information about the tagging shooter. This information will be processed by the tagged vest, and relayed to the base station.

In addition to filtering incoming signals from the trigger mechanisms, the Digispark is also responsible for constructing the encoded data packet. The structure of the packet is based off of a common Sony television IR remote and its technology [5]. Typically, a pulse width modulated signal

sends the bits to the receiver. Based on the length of a HIGH pulse, a 1 or 0 is sent while separated by a LOW pulse of a fixed length.

A single pulse is comprised of the following:

- A header
- The information to send
- A parity bit

In the case of the Tagged glove, the information being sent by the laser pulse would be the shooting player's *Player ID*. During game set up, Player IDs are distributed amongst the players via distribution of the laser-shooting glove. This will serve as the basis of scorekeeping and for users to check their statistics. For the sake of the prototype, each player's glove is hardcoded with a Player ID. In **Table 5**, a sample table of four players and their corresponding Player IDs are shown. Based off of the available bits, the Player ID assignment can be designed for up to 16 players if the project was to be scaled up.

Table 5: Sample of Player Numbers and Their Corresponding Player ID

Player #	Player ID (bits)
1	0000
2	0001
3	0010
4	0011

Similar to the construction of the data packet, the header is represented by a HIGH pulse with a length of 2400 μs . A 1200 μs HIGH signal characterizes a "1", while a 600 μs HIGH signal denotes a "0". A 600 μs of LOW signals separates each HIGH pulse. The header and parity bit wraps the Player ID, as it is the information that will have to be sent, and an example of what is sent can be found in **Figure 6**. The parity of the data packet adheres to even parity and provides a source to check the validity of the information transmitted. If there is an even number of 1's being sent, the parity bit will be 0, whereas an odd number of 1's will yield a parity bit of 1.

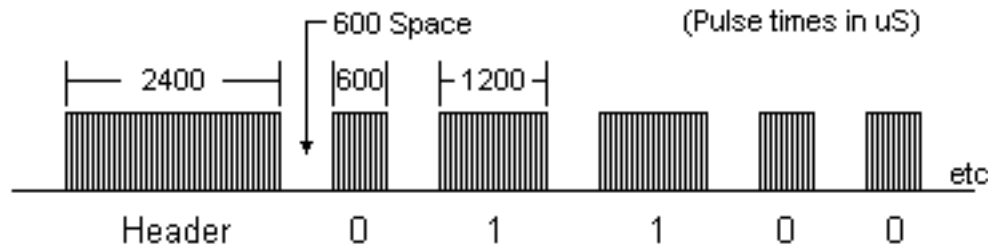


Figure 6: Encoded Laser Pulse [6]

A complete data packet takes a minimum of 9000 μs and a maximum of 10 800 μs to transmit. This works out to 92 shots per second at most, which is much faster than the 190 ms for average human reflexes to react to visual stimulus [7] and trigger a shot. This works well with the trigger design as it provides a large margin in latency and meets the design specifications of accurately detecting 90% trigger actions.

3.2 Vest Subsystem Design

At a holistic level, the vest serves the purpose of being the receiver of the laser. It hosts the components of the system responsible for detecting whether a player was hit, and communicating that information to the base station.

Six independent fibre optic patches span different areas of the vest, each relaying information regarding whether a player has been tagged to the onboard Arduino. The Arduino board is responsible for processing incoming data (a shot), and resultantly providing haptic feedback to the user via the use of vibration motors mounted between the vest and fibre optic patch. Additionally, it is responsible for communicating the shot it detected to central game hub (base station).

In addition to fulfilling the technical specifications of the project, the vest must also be comfortable enough for players to run around in without discomfort and chafing. The cut of the vest and inclusion of adjustable belt straps aligns with the need to fit various body types and meets typical US unisex sizes (**see Appendix A**). The material of the vest is akin to canvas, similar to those commonly used for outdoor recreational purposes and combat fatigues, to provide some durability and structure. It is expected that additional changes to the vest's form may have to be made once the vest has been tested in a practical setting and in real world conditions.

An electronics pouch houses the electrical components of the vest subsystem. The pouch is situated

in the lumbar region of the player's lower back, as its placement there causes the least obstruction to the players when they are running and hiding.

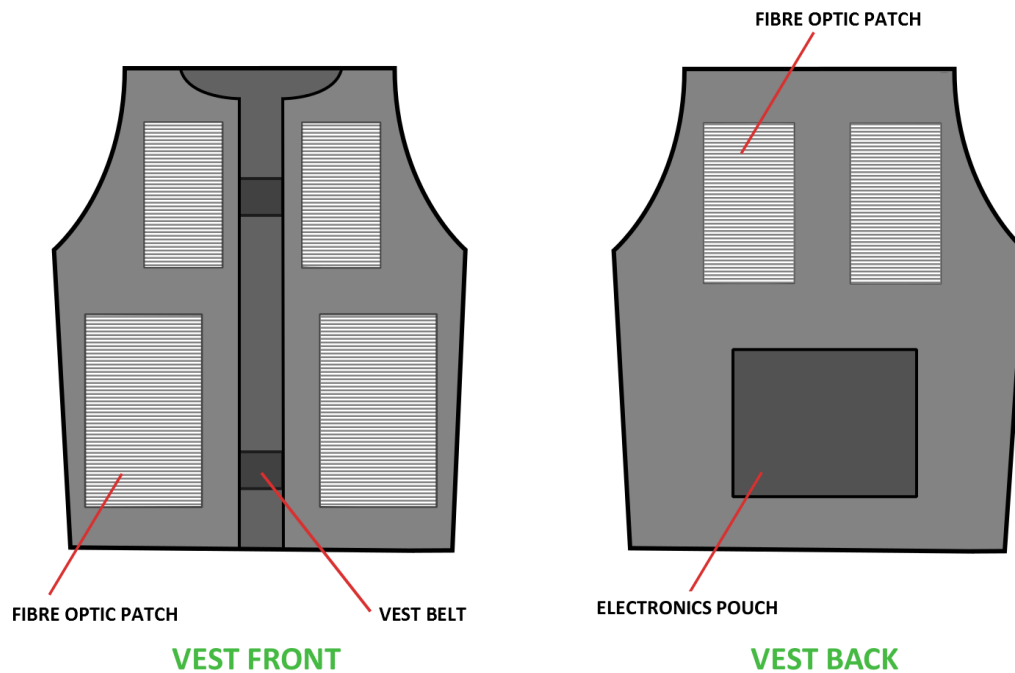


Figure 7: Final Vest Layout

In **Figure 7**, the diagram of the vest design shows a possible arrangement of six sensor patches that a player can tag on another player's vest. Each of these patches have a 3-bit identifier assignment and a pre-determined amount of points awarded for a successful tagging (as listed in **Table 6**).

Table 6: A Listing of Sensor Patches, the Corresponding Patch ID and Associated Points

Patch (as shown in Figure 7*)	Patch ID (bits)	Points
Front, Top-Left	000	2
Front, Top-Right	001	4
Front, Bottom-Left	010	1
Front, Bottom-Right	011	1
Back, Top-Left	100	2
Back, Top-Right	101	2

*Facing the reader as is

3.2.1 IR Laser Detection Mechanism

The Tagged system relies on the effectiveness of an IR sensor that is thin and flexible enough to be worn. As such, prior to arriving to the current design, a variety of options were explored in tandem with consultations made with professors who specialize in flexible light sensors.

During the earlier research stages of the project, an interesting use of “LightCloth” was identified, in which it served as the basis for a research project aimed at creating interactive surfaces through fabric. “LightCloth”, as it turns out, has diffusive fibre optic cables interwoven along a single plane of the fabric, providing a bidirectional channel for information to travel through. Due to this property, fibre optic fabric is a great medium for light conduction and fulfills the project’s need for a thin and wearable sensor.

Some obstacles arose in trying to obtain fibre optic fabric samples that fit within the group’s budget, leading to the investigation of alternatives. After engaging in conversations with Professor William Wong, Professor Simarjeet Saini and Professor Dayan Ban, it was deemed that amorphous silicon and amorphous germanium sensors would be viable options. Unfortunately, if this option were to be exercised, the vest portion of the system would have to be scaled down from a practical product to a proof-of-concept approach, if only to illustrate the potential of using cutting-edge flexible light sensors instead. After further deliberation, it was noted that logistical issues such as accessibility and availability to these still-in-research sensors would greatly hinder the ability to meet deadlines in a punctual manner.

After determining that the amorphous sensors would not be a viable approach, sheets of fibre optic fabric were obtained. Some initial testing was done to confirm the feasibility of using fibre optic fabric to conduct light from a transverse source. The testing was done with an Near IR (NIR) laser rated at a wavelength of 850 nm and a photodiode whose peak performance corresponds to 850 nm, as well. Testing data can be found in **Appendix B**.

3.2.2 Current Design

From the data collected, one can observe that the fibre optic fabric performs best when doubled up, with the laser diode having a 45° angle of incidence with respect to the direction of the fibre length, and under dim lighting conditions. The more light that enters the fibre optic length from the horizontal, the better the reading, since the fibre optic patches on the vest are to be placed in such a manner that their grain runs horizontally. This design choice was made based on the expectation

that there will be only a small number of laser pulses directly aimed from above or below a player. This implies that it is best to have the fibre optic lines running across the player's body as this orientation maximizes the amount of Near IR light transmitted from direct or side shots.

Additionally, an effort to include a visible light filter is being made to improve readings and reach a consistent value under most lighting conditions. Using spotlight colour filter gel sheets, a visible light filter can be constructed and mounted in front of fibre optic fabric. This filter blocks out most light in the visible light spectrum, while allowing NIR and IR light to pass through.

Each fibre optic patch has a NIR photodiode attached to the bundled fibre optic ends to detect laser pulses. Once a valid response is received, the Arduino sends haptic feedback to the vest via the vibration motors mounted in between the fibre optic patches and the vest. This indicates to the player that they have been tagged. Meanwhile, the vest's Arduino board decodes the laser pulse and the information is sent off to the central hub for scorekeeping.

3.2.3 Communication Protocol

Subsequent to the decoding of the laser pulse once hit, the vest relays the data back to the base station. To accomplish this, the XBee modules used in each vest serves as nodes of a star network, while the central hub acts as a central node.

The XBee module was selected for its ease of use and low power consumption. Further analysis and details of this decision can be found in section 3.3.2. Communicating using the ZigBee protocols, which builds upon the IEEE 802.15.4 protocols, also allows for flexibility in setting up a variety of network topologies. In particular, much of the network setup, including seeking, bridging connections to other nodes and self-healing properties, are already built into the module and provides the reliability and robustness required for the project. As well, there are a vast number of documented tutorials available online for prototyping wireless communication between devices, which provides ease of setup and integration.

Once the XBees are connected and a part of the network, information can be sent in a serial fashion between two nodes. The information is bundled as a data packet. Unlike the laser pulses sent by the glove, the Xbee modules come with a variety of options such as *packet retries* that ensure the receipt of the data packets. As such, there is no need for a header. The information that must be sent to the central hub includes the player that was tagged, the tagging player and the area (patch) that was tagged. The central hub processes the information and rewards points to the appropriate player.

The structure of the vest-to-hub data packet which will be sent is as follows:

[Tagged Player ID] [Patch ID] [Tagging Player ID]

where

- *Tagged Player ID* is the 4-bit identifier of the player who had been tagged
- *Patch ID* is the 3-bit identifier of which sensor patch had been tagged
- *Tagging Player ID* is the 4-bit identifier of the player who did the tagging

As an example, the following code is used to send a packet of data across the ZigBee network, and would be included in the Arduino functionality:

```
// Scenario: Player 1 gets hit by player 3 at the front, bottom left patch
XBeeAddress16 address = new XBeeAddress16(0x22, 0x22); // address for base station
int payload[] = { 1, 2 , 3}; // tagged player id, patch id, tagging player id
TxRequest64 packet_tx = new TxRequest64(address, payload);
xbee.send(packet_tx); // send packet to base station
```

3.3 Base Station Subsystem Design

The base station serves two main purposes: to be able to communicate with all the vests on the field in order to log scoring during each game, and to be able to parse and display results from a played game to an easily accessible website. Other subtasks which the base station takes care of include the signalling the start and end of each game, tracking the timing for the game, and also ensuring that the vests are properly signalled upon a player leaving the bounds of the arena. Thus, there are three key areas of design for the central hub subsystem - deciding on the main microcontroller, deciding on which wireless technology and associated module, and finally, the design of the game-results application.

3.3.1 Deciding on the Board

The microcontroller chosen to design the base station around must be able integrate with any necessary modules, efficient in power consumption, and capable of running essential programs and applications. These general necessities are based off of the associated project specifications; in particular, the ability for wireless communication with the vest, the battery life being longer than 1.5 hours, and the ability for viewing of in-game stats through a smartphone application.

Allowing for the microcontroller board to host a web application which can be accessible through a smartphone has a few key prerequisites - the hosting of a WiFi network for the smartphone to

connect to, and an operating system to allow for application hosting and any necessary processing. For the operating system, Linux was chosen, as it provides a suitable environment for data processing, has many libraries already encapsulated (such that extra time would not be taken in order to define the network protocol libraries by hand) and it can also run technologies such as Python/HTML5 for hosting the phone applications. The design of the application itself is described in Section 3.3.3 of this document.

Based on the above discussion, the ideal microcontroller for the base station one that:

- Either has a Wifi module integrated, or has the potential for integration
- Runs Linux
- Has a current draw below 500 mA

Using the above criteria as a starting point, the following design iterations were considered and evaluated accordingly.

3.3.1.1 Iteration One - Raspberry Pi + WiFi Module

The first design iteration chosen was to take a Raspberry Pi microcontroller and integrate a WiFi module onto it. The Raspberry Pi Model A, in particular, was chosen because it runs Linux, and has a current draw which can go as low as 500 mA [8]. Integration of the WiFi module with the board would include the purchase of a WiFi shield for the Raspberry Pi, the connection of the two devices, and then testing. Another necessary requirement of this approach would be to find and download, or program, libraries to the board which are necessary for being able to send communications between devices according to the 802.11 standard.

Though this solution covers all of the technical requirements, the addition of time constraints from the initial prototype make the approach less appealing. On top of needing to wait for multiple products to ship, the finished microcontroller will arrive in separate pieces. The solution requires extra time for both integration as well as testing - which is not feasible given the project deadlines.

3.3.1.2 Iteration Two - Arduino Yún

To solve the time constraint issue, the solution of purchasing the Arduino Yún was considered. Not only does the Arduino Yún host a Linux operating system *in addition* to hosting an Arduino processor, it also has a WiFi module already integrated. Finally, this solution also has a current draw which meet the requirements set - with only 315 mA of current being drawn as a maximum [9].

The Arduino Yún saves time by having better bridging capabilities between the Arduino processor and Linux operating system. Additionally, it has seamless integration with a built-in Wifi module, which includes WiFi libraries already on the board.

3.3.2 Communication between Vests and the Base Station

Many essential functional specifications of Tagged involve the ability of the base station to communicate with vests which are out on the playing field. For one, the base station must be capable of wireless communications with vests in order to receive hit information throughout the course of a game. It will also require wireless capabilities in order to signal to the vests when the start and end of a game occurs. Furthermore, there are the requirements on the chosen approach to have a communication range of at least 85 ft, and the battery life lasting longer than 1.5 hours between recharges. Based on these particular specifications, three choices for wireless technologies were decided upon for further analysis - WiFi, Bluetooth, and ZigBee.

The particular modules which were chosen for further comparison and analysis are the WiFi Shield v2.1 for Arduino as the WiFi module, the XBee Series 1 for ZigBee, and the Bluetooth SMD module - RN-42 for Bluetooth. These particular product choices were decided upon based on how well they could potentially satisfy the specifications. Note that this solution will require integration on both the vest side as well as the base station side. Therefore, though the base station board has WiFi already integrated, it will also require integration into the vest subsystem's Arduino board.

3.3.2.1 Bluetooth

Many essential functional specifications of Tagged involve the ability of the base station to communicate with vests which are out on the playing field. For one, the base station must be capable of wireless communications with vests in order to receive hit information throughout the course of a game. It will also require wireless capabilities in order to signal to the vests when the start and end of a game occurs. Furthermore, there are the requirements on the chosen approach to have a communication range of at least 85 ft, and the battery life lasting longer than 1.5 hours between recharges. Based on these particular specifications, three choices for wireless technologies were decided upon for further analysis - WiFi, Bluetooth, and ZigBee.

The particular modules which were chosen for further comparison and analysis are the WiFi Shield v2.1 for Arduino as the WiFi module, the XBee Series 1 for ZigBee, and the Bluetooth SMD module - RN-42 for Bluetooth. These particular product choices were decided upon based on how well they could potentially satisfy the specifications. Note that this solution will require integration on both the vest side as well as the base station side. Therefore, though the base station board has WiFi already integrated, it will also require integration into the vest subsystem's Arduino board.

3.3.2.2 WiFi versus ZigBee

The remaining options for wireless technologies and associated protocols are the WiFi option and the ZigBee option - or the Arduino WiFi Shield and the ZigBee Series 1 respectively. Based on the related project specifications listed in section 3.2.2 above, the following 6 criteria were chosen to help decide which option is more suitable for the Tagged system: transmission distance, power

consumption, ease of integration, cost, availability, and scalability. As a note, reliability of the technology is also an important consideration, however, both technologies are nearly equal on this front, so it has been left out of the decision matrix criteria listing. The transmission distance for both remaining options are above 85 feet. The preferred option will have a high transmission distance, low power consumption, be easy to integrate, will be readily available from Canadian providers, and will be low in cost. Finally, the scalability of the chosen technology will be taken into account for better future development of the Tagged system. The decision criteria, associated weightings and then the score given are listed in **Table 7**, with the scores explained below.

Table 7: Decision Matrix for Deciding Wireless Communication Technology

Criteria	Weight (%)	Option 1: Arduino WiFi Shield / WiFi	Option 2: ZigBee Series 1 / ZigBee
Transmission Distance	10	10	10
Power Consumption	20	3	10
Ease of Integration	30	8	9
Availability	15	7	10
Cost	15	4	7
Scalability	10	5	8
Total	100	6.15	9.05

In regards to the transmission distance, both technologies [11] [12] provide a range of around 100 to 150 feet in an indoor setting, and a range of around 300 feet in an outdoor setting. Thus, both technologies satisfy the project specification relating to the necessary range. Also, because both values are similar, and more than sufficient, both WiFi and ZigBee were given a score of 10 for the transmission distance criteria. The power consumption of the ZigBee series 1 module is very low, at only 1 mW during transmissions [12], while the power consumption of the WiFi shield is up around 130 mW on average [13]. Since power consumption is preferred to be lower, the score of 10 will be given to XBee module, and a 3 will be given to the WiFi module.

As there are a large number of tutorials online which provide details on implementation, and there is prior evidence which demonstrates that integration of the chosen Arduino boards and the XBee module is possible, the ease of integration for the XBee module receives a score of 9. There are fewer resources online which provide aid in regards to implementation of the integration between the WiFi shield. However, since the chosen microcontroller for the base station already has the WiFi module integrated, integration would only require working the module into the Arduino board for the vest subsystem. Therefore, the score of 8 shall be given to the WiFi module.

Availability of the XBee module is high, because they are both available through the ECE store, and able to be purchased online from Canadian distributors. Since there were a few modules available through the ECE store, the cost is also considerably low. Normally, however, the cost of an XBee series 1 module is \$21.96 [14], which will be the cost taken into account for the scoring, as extra modules will need to be purchased in order to fulfill the wireless requirements of all devices. The WiFi shield v2.1 for Arduino modules are available through Canadian distributors, however, there are few options, and availability from said distributors is low. This being said, the price of the Arduino WiFi shields are also significantly higher - being generally in the range of \$94.44 per module[15]. Thus, the scores were decided as they have been shown in **Table 7**, above.

Finally, in regards to scalability, the provided scores are due to the fact that WiFi, in general, consumes more power and requires a more expensive power source. Additionally, the module itself is more expensive. If Tagged were to be mass produced, the extra costs of these components would make the project less scalable. The ZigBee modules are cheaper, with lower power consumption - so mass production would benefit from these traits. Also, the ZigBee networks are generally better at providing personal networks, with various network configurations possible.

3.3.2.3 Final Decision

Based on the total scores calculated in **Table 7**, the chosen approach is to use the XBee Series 1 modules, with ZigBee communications for both the vest and the base station. The ZigBee option scored a fair bit higher than the WiFi option, with a final score of 9.05. This is primarily due to the fact that the WiFi modules tend require higher power in order to transmit, and ended up being more expensive. This solution satisfies the specifications regarding wireless communications between the vest and central hub, having a range more than 85 feet, and also having a low enough power consumption to aid in fulfilling the battery life requirements.

3.3.3 Parse and Display Results

In order to store results, rather than using an entire database, memory on the Arduino will be saved by keeping the game data in simple arrays within the local heap. This means that game results will currently remain transient, however, future designs might include long-term storage of results. As it currently stands, however, a simple web portal with a backend to compute the scores throughout the game and a frontend to display the results is all that is needed.

- The construction of the web portal will be done using the Python scripting language (which is natively installed on the OpenWRT distribution included with the Arduino Yún [16]) in addition to the use of a web framework, which will enable faster development of the portal due to templating. Since the Arduino Yún has a limited amount of memory, 32 KB in total with 4KB dedicated to the bootloader, most web frameworks need to be avoided since they

take up a considerable portion of the allocated memory. As such, the Python microframework *Flask* was deemed the natural choice to complement the use of python on the Yún. In general, the web application portion shall be divided into the following sections:

-
- Backend - helper methods to parse the data and perform computations
 - Python (with the help of the Python Flask framework)
- Frontend - displaying the results in a neat and easy to understand table
 - HTML and CSS
 - Minimal Python (to pass in the relevant information to be displayed)

As mentioned earlier, since Tagged is not designed to maintain persistent score data of previous games, it does not need to make use of a database. Instead, during game-play, a simple array suffices in keeping track of the players' score. Scores are calculated based on the number of hits a player has taken (their damage) and the number of tags they have made (the number of times they have made a successful shot). With respect to the array used to store score information, the player's ID corresponds to a particular index of the array. For example, the value at index 1 of the array indicates the player 1's score, while the value at index 8 indicates the number player 8's score. Every time a player's vest detects a hit, the XBee module in the vest will communicate that to the base station. Within the base station, this information will then be communicated from the Arduino portion of the Arduino Yún, over to the web application that lives within the Linux distribution portion. Once this information is parsed by the backend of the web application, the array which stores the score information will be updated appropriately, and the web page will reflect the score at the instant in time it is accessed.

An example of pseudocode that summarizes the Arduino Yún parsing the XBee data packet within the Arduino partition, and then placing it in a common buffer for the Linux partition to access is given below:

```
// store parsed packet information, and then access it easily in other
sections of code

typedef struct{
    int tagged_player_id;
    int patch_id;
    int tagging_player_id;
}TagInfo;

TagInfo parsed_packet; // instance of struct data type TagInfo
```

```

int main (void ){ // main method

...

    // if serial data available to read then grab the relevant information
    if(Serial.available() > 0){

        // first read is to get tagged player's id
        parsed_packet.tagged_player_id = (int)Serial.read();

        // second read is to get the patch id
        parsed_packet.patch_id = (int)Serial.read();

        // third read is to get the tagging player's id
        parsed_packet.tagging_player_id = (int)Serial.read();

    }

...

    Bridge.put(1, parsed_packet.tagged_player_id);           // place value
into buffer

    Bridge.put(2, parsed_packet.patch_id);                   // place value
into buffer

    Bridge.put(3, parsed_packet.tagging_player_id);          // place value
into buffer

}

```

The Arduino Yún's built in bridge library will be used in order to share information between the Arduino and Linux processors. In particular, calls such as `Bridge.put(key, value)` shall be used to place the struct data from the ATmega32U4, into an intermediate buffer which can be accessed by the Atheros 9331 processor running Linux. This information can then be retrieved from the intermediate buffers using `Bridge.get(key, buffer, buffer_length)` On the Linux side (similar to how it was placed into the buffer), and can appropriately be handled by a helper library within the web application. The helper library will appropriately separate out and assign the values of the `tagging_player_id` and `tagged_player_id` to variables. Additionally, it will determine the associated score-value to the patch that was hit based on **Table 6** in section 3.2. This score-value will then be used to deduct from the `tagged_player_id`'S SCORE, and add to the `tagging_player_id`'S SCORE .

An example of the web application's helper function to update the array of player scores is given below:

```

int player_scores []; // initialize and set array to keep track of player scores

```

```

int score; // variable to hold actual score value based on patch id
...

// calculate the score based on the patch id and assign this to the variable score
...

// update the scores of the person who *was* tagged, and the person who did the
tagging

void update_score (int tagged_player_id, int score, int tagging_player_id){
    player_score[tagged_player_id] = player_score[player_id] - score;
    player_score[tagging_player_id] = player_score[player_id] + score;
}

```

3.3.4 Final Base Station Design

To reiterate what was stated in the beginnings of section 3.3, there are three key areas of design within the base station subsystem - the decision behind which microcontroller to use for the device, the decision as to which wireless communications technology should be used, and finally, the design behind parsing and displaying game results.

In order to accomplish these necessary tasks, the Arduino Yún was chosen as the microcontroller to design the base station around, given the functional flexibility it offers as a monolithic piece of hardware. The Yún contains two processors that are able to communicate with each other seamlessly - the ATmega32u4 for regular Arduino functionality, and the Atheros AR9331 to run the OpenWRT Linux distribution.

The Yún board will be wired up to three other components - the XBee Series 1 module for communication with the vest, a LiPo battery that will satisfy the battery life requirements, and finally a simple button. The button will be used to trigger the start of a game, where subsequent to the button press, a 30 second countdown will be activated to provide the players sufficient time to scatter. They will be alerted that the 30 second countdown is ongoing by the central hub signaling their vests to continuously vibrate for the countdown duration. Once the 30 seconds have elapsed, the game will begin, with associated scorekeeping, wireless transmissions and game duration

countdown being handled by code running on the Arduino Yún microcontroller. When a timeout has occurred, another wave of wireless signals will be sent to the vests through the XBee modules to cause a continuous vibration in order to tell the players that the game has finished.

4 Discussion and Project Timeline

4.1 Discussion on Design

The hours of work completed by each student can be found in Table X below.

Table 8: Hours of Work Put in by Team Members

Student	Hours of Work
(names removed)	82.5
	84.0
	79.5

As can be seen from the table, the amounts of time which has been put into working on Tagged over the past 9 weeks of school have not necessarily been the ideal 10 hours of work per week. However, this was primarily due to the fact that parts were needing to be ordered in, and there was an associated delay in receiving them. The present scheduling takes this into account, and demonstrates that 50% of the initial prototype is a realistic goal.

Tagged, as a project, draws on many facets of Electrical and Computer Engineering. The vest and base station subsystems, though basic in their circuitry, use ECE 358 and ECE 418 knowledge on communication protocols and concepts of networking in order to successfully relay scoring information. Analog and digital signal processing knowledge from both ECE 318 and ECE 413 is required when working to improve the efficacy of sensors readings, and also aided in the determination of the laser encoding scheme (see section 3.1.4). Finally, as all three subsystems of Tagged are embedded systems, involving the integration of both hardware and software, knowledge from ECE 455 and ECE 423 is and has been highly applicable.

Despite the major changes made to the glove subsystem due to the inadequate flex sensors' performance, the group is more than pleased with the current design and believes that it should be sufficient in satisfying the required triggering functionality. With the triggering mechanism as specified and the microcontroller being small and having low power consumption, all of the project specifications associated to the glove subsystem have been met by the final design.

The trigger detection mechanism is much simpler yet more elegant than the initial designs, with the capacitive touch property of the conductive fabric offering a great improvement. There is no additional

filtering of noisy signals necessary, and the conductive fabric design offers no perceivable latency, and improved accuracy over the initial design. The glove subsystem no longer relies on a physical sensor that may not give consistent results between trials, and will not be damaged when creased. Additionally, the use of capacitive regions removes the need to calibrate onboard sensors prior to the start of each game. In terms of future refinement for this subsystem, additional effort will be put in to improve the aesthetics of the glove.

The vest subsystem design not only satisfies all the essential, functional and non-functional specifications, as outlined in Section 2, the construction of the vest plans on satisfying the non-essential non-functional specifications of being lightweight, durable and offer a large target coverage for users to tag. Additional testing must be completed, but the group is confident that all associated specifications are both reasonable and attainable.

In particular, the group is quite proud of the creative approach taken in fashioning a flexible, wearable medium as an interface for data transmission. Further work needs to be done in order to refine the IR sensor patch such that performance is consistent in reasonable indoor and possibly outdoor lighting situations. The group feels that this issue is the largest hurdle of the Tagged project and once resolved, will allow for work to scale up and completed more quickly.

As can be seen in Section 3.3, the central hub subsystem relies as much on the hardware as it does on the software. Discovering the Arduino Yún was a large accomplishment for the project. The WiFi capabilities and the addition of a Linux environment helped glossed over much of the lower level communication setup which would have been required. Instead, the group is able to start prototyping much sooner. Now that the hardware components are chosen for subsystem and satisfies all the physical functionalities of the base station, the remaining focus is on the software side of the device. As software development is highly iterative process, the design in this area will continually have refinements being made to it. It is likely that the group will run into obstacles working with the Linux bridging properties of the board and will have to resolve them creatively.

The likely hazards of the decided upon solution which may occur during the construction phases of the prototype involve any hazards introduced by the circuitry being designed and built by the group, the core reliance upon unprotected lasers of Class 1M or stronger being shot at other individuals, and then any dangers associated with the electronics being attached to clothing which will be worn by a human being. For one, all circuitry involved will be designed as specified by this document, and then built by the group members. Thus, safety considerations must be taken into account while completing the circuitry involved would especially include having a proper encasing for all microcontrollers and power sources. Since Tagged is a game system, it would involve individuals running around with the equipment strapped to them - which would imply safety associated with the circuitry is key.

4.2 Project Timeline

Assuming each day of work consists of an individual working for 8 solid hours on a particular task, then the following table outlines the timeline for the remainder of the 4A term as well as the schedule for the 4B term. In **Figure 8**, is a Gantt chart of the group's proposed timeline.

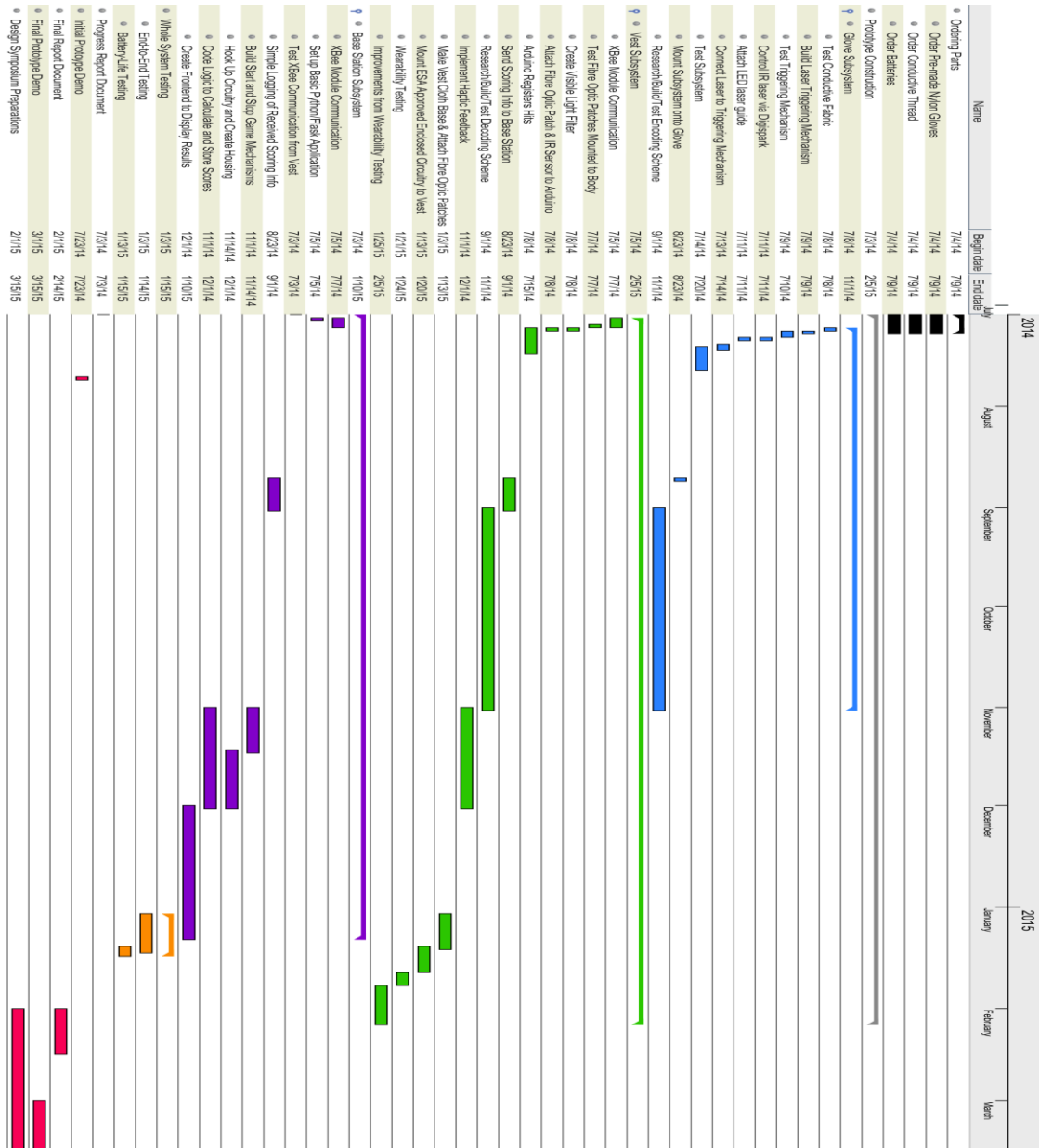


Figure 8: Group 51 Proposed Timeline

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Appendix A: US Unisex Standard Clothing Sizes



	XXS	XS	S	M	L	XL	2XL
Chest (inches)	29-31	30-32	34-36	38-40	42-44	46-48	48-50
Waist (inches)	27-29	28-30	30-32	32-33	33-34	36-38	40-42

Source: <https://www.americanapparel.net/sizing/default.asp?chart=mu.shirts>

Appendix B: Fibre Optic Fabric Test Data

Date: June 30, 2014		Photo Diode: SFH 203P		Laser Wavelength: 850 nm	
Diode Reading (1,2,3) - V	Laser On/Off	Lights On/Off	Angle wrt. Fabric	Double Layered	Reflective Backing
0.209 -	-	On	-	No	No
0.209 On	On	On	90	No	No
0.219 -	-	On	-	No	Yes
add 0.3 mV On	On	On	90	No	Yes
0.154 -	-	On	-	Yes	No
add 0.8 mV On	On	On	90	Yes	No
0.153 -	-	On	-	Yes	Yes
add 1.6 mV On	On	On	90	Yes	Yes
done -	-	On	-	No	No
add 1.3 mV On	On	On	45	No	No
done -	-	On	-	No	Yes
add 1.5 mV On	On	On	45	No	Yes
done -	-	On	-	Yes	No
add 2.5 mV On	On	On	45	Yes	No
done -	-	On	-	Yes	Yes
add 3.5 mV On	On	On	45	Yes	Yes
3.25 mV -	-	Off	-	No	No
-2.8 On	On	Off	90	No	No
3.39 mV -	-	Off	-	No	Yes
add 7 mV On	On	Off	90	No	Yes
1.53 mV -	-	Off	-	Yes	No
-4.22 On	On	Off	90	Yes	No
1.61 mV -	-	Off	-	Yes	Yes
-10.93 On	On	Off	90	Yes	Yes
done -	-	Off	-	No	No
-14.7 On	On	Off	45	No	No
done -	-	Off	-	No	Yes
-18.31 On	On	Off	45	No	Yes
done -	-	Off	-	Yes	No
-9.51 On	On	Off	45	Yes	No
done -	-	Off	-	Yes	Yes
-12.58 On	On	Off	45	Yes	Yes